Reyhane Javidi Alsaadi

Computer Engineering Scholar | Vision Foundation Models & Edge AI Researcher

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Final-semester Computer Engineering undergraduate at Dr. Shariaty Technical University (**GPA: 19.16/20 | ~3.97/4.0, Top 5% of Cohor**t). Research focuses on **Vision Foundation Models (DINOv2, BioCLIP)**, **Few-Shot Learning,** and resource-constrained **Edge AI.** First and corresponding author of the **BioDINO-Edge** framework (+13.0% over SOTA on in-the-wild plant pathology benchmarks). Seeking **M.Sc. / Ph.D. Graduate Research Assistantship** positions for Spring 2027 / Fall 2027.

Skills

Artificial Intelligence & Vision: PyTorch, TorchVision, Timm, Scikit-Learn, NumPy, SciPy, Vision Foundation Models (DINOv2, BioCLIP), Few-Shot Learning, Regularized Fisher-LDA

Edge AI & Low-Latency Deployment: ONNX Runtime, INT8 Dynamic Quantization, Hardware-Accelerated Inference, Low-Latency Memory Pipelines, Python

Scientific Writing & Tooling: LaTeX (IEEEtran & Scientific Writing), Linux

Education

Bachelor of Science (B.Sc.) in Computer Engineering — Software 2024 – Present | Tehran, Iran

Dr. Shariaty Technical University, Tehran, Iran

Final-Semester Senior (Expected 2026) | **Cumulative GPA: 19.16 / 20.00 (~3.97 / 4.00)** — **Top 5% Departmental Distinction**.

Selected Coursework: Algorithm Design (20/20), Artificial Intelligence (20/20), AI Lab (20/20), Theory of Automata (20/20), Software Engineering (20/20), Probability & Statistics (19.5/20).

Associate of Science (A.Sc.) in Computer Software Applications 2022 – 2024 | Tehran, Iran

Dr. Shariaty Technical University, Tehran, Iran

Cumulative GPA: 18.96 / 20.00 (~3.91 / 4.00) — Top 5% Departmental Distinction.

Selected Coursework: Data Structures (20/20), Advanced Programming (20/20), Database Systems (20/20).

Publications

BioDINO-Edge: Biologically-Grounded Multi-Scale Vision Foundation Models for Trustworthy Few-Shot Plant Pathology on Resource-Constrained Hardware, *Submitted to Computers and Electronics in Agriculture (CompAg) [14-page IEEEtran Format, 2026] — Preprint available upon request.*

- Proposed a dual-foundation feature representation **(896-d)** combining spatial patch descriptors from **DINOv2** with taxonomic priors from **BioCLIP** for few-shot plant pathology on edge devices.
- Formulated a closed-form **Fisher-LDA** alignment using analytical **Ledoit-Wolf covariance shrinkage**, bypassing backpropagation on singular few-shot matrices **(9.8K FLOPs, <0.003 ms latency).**
- Reached **98.72% peak accuracy** on PlantVillage and **83.61% (+13.0% over SOTA)** on the unconstrained PlantDoc benchmark, while reducing calibration error (ECE) from **37.97% to 0.81%** via temperature scaling.

Projects

Edge AI Model Deployment & Optimization (BioDINO-Edge PoC), *SysML & Edge Inference Prototyping* 2026

- *Focus: Model Quantization, ONNX Runtime Mobile, Hardware Benchmarking*
- Quantized the PyTorch dual-foundation vision backbone into an **INT8 ONNX model** (22 MB footprint) for resource-constrained edge execution.
- Built an offline on-device inference prototype via **ONNX Runtime Mobile (NNAPI)** to evaluate live latency and throughput on mobile ARM processors.
- Implemented mathematical pre-inference guardrails (luminance thresholding & Shannon margin-entropy confidence gating) to reject out-of-distribution inputs.

Teaching Experience

Teaching Assistant (TA) — Game Development 01/2026 – 07/2026

Department of Computer Engineering, Dr. Shariaty Technical University

Conducted weekly laboratory sessions and technical office hours for **40+ undergraduate engineering students** on interactive graphics pipelines, physics simulations, and state-machine logic.

Honors & Awards

Presidential Letter of Commendation (Ref: 5039/202/1615), *Dr. Shariaty Technical University* 2026

Conferred by the **University President** in recognition of outstanding **instructional dedication**, peer mentorship, and faculty collaboration in the Department of Computer Engineering.

Top 5% Scholastic Distinction (Dean's Honor List), *Dr. Shariaty Technical University* 2026

Conferred for maintaining **consecutive top-tier departmental ranking** across both B.Sc. and A.Sc. degree programs.

Prospective Graduate Research Interests

- **Few-Shot Learning & Foundation Models:** Multi-scale feature alignment, parameter-efficient adaptation, and representation learning for low-data regimes.
- **Resource-Constrained Edge AI:** Closed-form analytical discriminants, post-training model quantization (INT8/INT4), and zero-gradient on-device learning.
- **Trustworthy & Calibrated AI:** Uncertainty quantification, conformal prediction, and reducing calibration error (ECE) in safety-critical automated diagnosis.

Languages

English: Professional Working & Academic Proficiency

Persian: Native